

Coding Human Health AMR laboratory function competency

Using the Fleming Fund 'Site Planning and Monitoring Forms'

This guide demonstrates how Fleming Fund site form questions are used to code the level of competency of AMR sentinel site functions for each laboratory (Supplementary Figure 1).

By following this guide, one can better understand the question answers and how they affect functional competency.

Component 1: Clinical admission assessment and investigation

Subcomponent 1a: Clinical admission assessment - Questions 1a1-1a3

Core

- To achieve 'Core', clinical history and examination and investigation must be based on physician (syndromic) diagnosis.
- As such, sites must answer 'Yes' to question 1a1:
 - 'Does the site have guidelines to help clinicians decide which microbiology samples to take from patients?'
- This answer demonstrates that there exist guidelines to help clinicians collect appropriate information, leading them to a syndromic diagnosis and assisting in microbiology sample selection. This ensures the process, and there the diagnoses, are consistent.
 - Whilst not explicitly stating that there must be collection of clinical history and an examination, this naturally forms part of the clinical investigative process.

Extended

- To achieve 'Extended', systematic clinical history and examination must be performed according to clinical algorithms in all patients presenting with suspected infection.
- As such, as well as 'Yes' to question 1a1, sites must also answer 'Yes' to question 1a2:
 - 'Do the guidelines include standardised algorithms to ensure consistency?'
- Answering 'Yes' demonstrates the implementation of a standard clinical algorithm in patient clinical engagement, ensuring standardisation across sites.

Advanced

- To achieve 'Advanced', standardised admission proforma documenting clinical signs and symptoms must be used to guide diagnosis.
- As such, as well as 'Yes' to questions 1a1 and 1a2, sites must also answer 'Yes' to 1a3:

- 'Is patient sampling guided by a standardised admission proforma, including clinical signs and symptoms?'
- Answering 'Yes' demonstrates that a sufficiently comprehensive proforma (including signs and symptoms) is implemented during patient sampling, which forms part of the diagnosis pipeline.

Question	Core	Extended	Advanced
1a1	Yes	Yes	Yes
1a2	No/In progress/Not applicable	Yes	Yes
1a3	No/In progress/Not applicable	No/In progress/Not applicable	Yes

Subcomponent 1b: Clinical data - Questions 1b1 and 1b5

Core

- To achieve 'Core', clinical data must be included in (paper) requests for laboratory investigation, with unique alphanumeric identifiers.
- As such, sites must answer 'Yes' to question 1b1:
 - 'Is a unique alphanumeric identifier (ID number) assigned to patients and so that laboratory results can be linked with the correct patient?'
- They must also answer 'Clinical data included in paper request for laboratory investigation, with unique identifier' to question 1b5:
 - 'How are clinical/patient data included in requests to the microbiology laboratory?'
 - 'Other' is also considered an acceptable answer, which assumes the answer indicates that there is a form of clinical data inclusion with a unique identifier.
- Together, these answers demonstrate clinical data is collected on the laboratory requests and include a unique identifier.

Extended

- To achieve 'Extended', clinical data must be included in (electronic) requests for laboratory investigation, with unique alphanumeric identifiers.
- As such, as well as 'Yes' to question 1b1, sites must also answer 'Clinical data included in electronic request for laboratory investigation, with unique identifier' to question 1b5.

Advanced

- To achieve 'Advanced', the site must demonstrate linkage of extended clinical data (e.g. vital signs, blood results, outcomes) with laboratory data.
- As such, as well as 'Yes' to questions 1b1, sites must also answer 'Linkage of extended clinical data (e.g. vital signs, blood results, outcomes) with laboratory data' to 1b5.

Question	Core	Extended	Advanced
1b1	Yes	Yes	Yes
1b5	Clinical data included in paper request for laboratory investigation, with unique identifier/Other	Clinical data included in electronic request for laboratory investigation, with unique identifier	Linkage of extended clinical data (e.g. vital signs, blood results, outcomes) with laboratory data

Subcomponent 1c: Clinical investigation - Questions 1c1*

Core

- To achieve 'Core', systematic investigations must be based on physician syndromic diagnosis.
- As such, sites must answer 'Systematic investigation based on clinical findings' to question 1c1:
 - 'On what basis are clinical investigations done at the site?'
 - 'Other' is also considered an acceptable answer, which assumes the answer indicates that there is a form of systematic investigation performed.

Extended

- To achieve 'Extended', systematic investigation must be based on clinical findings.
- As such, sites must answer 'Systematic investigation based on clinician syndromic diagnosis' to question 1c1.

Question	Core	Extended
1c1	Systematic investigation based on clinical findings/Other	Systematic investigation based on clinician syndromic diagnosis

Subcomponent 1d: Training and quality assurance - Questions 2a1-2a3 and 2a12

Core

- To achieve 'Core', routine training must be provided for surveillance SOPs, quality control and Internal Quality Assessment.
- As such, sites must answer 'All staff' to question 2a1:
 - 'Have all relevant laboratory staff received up to date training for the AMR surveillance-specific standard operating procedures (SOP)?'
- Sites must also answer 'All staff' to question 2a2:
 - 'Have all relevant laboratory staff received up to date training on internal quality control processes?'
- This demonstrates that all relevant staff are receiving sufficient training on SOPS related to surveillance and maintaining quality.

Extended

- To achieve 'Extended', External Quality Assessment (EQA) must also be performed.
- As such, as well as 'All staff' to questions 2a1 and 2a2, sites must also answer 'Yes' to question 2a3:
 - 'Is the laboratory participating in an accredited External Quality Assessment scheme for bacterial identification and antimicrobial susceptibility testing?'

Advanced

- To achieve 'Advanced', the site must also function as a regional training centre.
- As such, as well as the above, sites must also answer 'Yes' to 2a12:
 - 'Does the laboratory provide EQA to other sites?'

Question	Core	Extended	Advanced
2a1	All staff	All staff	All staff
2a2	All staff	All staff	All staff
2a3	No/Not applicable	Yes, [EQA provider]	Yes, [EQA provider]
2a12	No/Not applicable	No/Not applicable	Yes

Component 2: Isolate identification and susceptibility testing

Subcomponent 2a: Sample transport - Questions 3a1-3a3

Core/Extended

- To achieve 'Core', samples must be transported according to local SOPs. To achieve 'Extended', samples must be transported according to international biosafety standards.
 - The questions in the HH site form ask about international biosafety standards, therefore 'Core' is skipped straight to 'Extended'
- To achieve 'Extended', sites must answer 'Yes' to question 3a1:
 - 'Does the site have SOPS for sample transport in line with international biosafety standards?'
- Sites must also answer 'All staff' to question 3a2:
 - 'Have all relevant laboratory staff received up to date training for the sample transport SOPs?'
- And finally, sites must answer 'Yes' to question 3a3:
 - 'Is the laboratory following the SOPs to ensure that samples are transported safely?'

Question	Extended
3a1	Yes
3a2	All staff
3a3	Yes

Subcomponent 2b: Sample registration - Questions 3b1

Core

- To achieve 'Core', sites must use a local laboratory paper-based data system.
- Sites must therefore answer 'Local laboratory paper based data system' to question 3b1:
 - 'How are clinical samples registered with the microbiology laboratory?'
 - 'Other' is also considered an acceptable answer, which assumes that a laboratory data system exists.

Extended

- To achieve 'Extended', sites must use an electronic laboratory data system.
- Sites must therefore answer 'Electronic laboratory data system' to question 3b1.

Question	Core	Extended
2b1	Local laboratory paper based data system	Electronic laboratory data system

Subcomponent 2c: Culture and identification - Questions 3c1, 3c7-3c14 and 3c17-3c20

Core

- To achieve 'Core', sites must use an automated blood culture system and the capacity to identify relevant priority pathogens according to SOPs.
- Sites must therefore answer 'Working automated blood culture instrument in place and under a service contract' to question 3c1:
 - 'Is the laboratory equipped with an automated blood culture system?'
- Sites must also answer 'Yes, according to SOP' to questions 3c7-3c14 and, if the site is answering after 2024**, questions 3c16-3c20.
 - 'Please indicate which of the following organisms the laboratory can process for identification and AST'

Required –

- 3c7: *Streptococcus pneumoniae*
- 3c8: *Staphylococcus aureus*
- 3c9: *Escherichia coli*
- 3c10: *Klebsiella pneumoniae*
- 3c11: *Acinetobacter* spp
- 3c12: *Salmonella* spp
- 3c13: *Shigella* spp
- 3c14: *Neisseria gonorrhoea*

Required 2024-onwards –

- 3c17: *Pseudomonas aeruginosa*
- 3c18: *Salmonella enterica* serovar Typhi
- 3c19: *Salmonella enterica* serovar Paratyphi A
- 3c20: *Neisseria meningitidis*

Extended

- To achieve 'Extended', sites must use an automated blood culture system, must be able to process Cerebrospinal fluid, urine, stool and swab cultures, and be capable of identifying all isolates according to SOPs for all priority pathogens.
- As well as the requirements for 'Core', sites must also answer 'Yes, according to SOP' to questions 3c2-3c6.
 - 'Please indicate which other samples the laboratory can process for bacterial culture, identification and susceptibility testing'

Sample types –

- 3c2: Cerebrospinal fluid
- 3c3: Urine
- 3c4: Stool
- 3c5: Pus/Pus swabs
- 3c6: Genitourinary samples for culture of *N. gonorrhoea*

Advanced

- To achieve 'Advanced', sites must perform automated identification (e.g. MALDI-TOF).
- Therefore, as well as the requirements for 'Core' and 'Extended', sites must also answer 'Yes' to question 3c15:
 - 'Is the laboratory using an automated system to perform bacterial identification?'

Question	Core	Extended	Advanced
3c1	Working automated blood culture instrument in place and under a service contract	Working automated blood culture instrument in place and under a service contract	Working automated blood culture instrument in place and under a service contract
3c2-3c14 (& 3c16-3c20 2024-onwards)	Yes, according to SOP	Yes, according to SOP	Yes, according to SOP
3c2-3c6	Yes, but without SOP/No	Yes, according to SOP	Yes, according to SOP
3c15	No/Not applicable	No/Not applicable	Yes

Subcomponent 2d: Susceptibility testing - Questions 3d1-3d2.1

Core

- To achieve 'Core', sites must use disc diffusion for blood culture priority pathogens according to SOPs.
- Therefore, sites must answer 'Disc diffusion' or 'MIC testing' to question 3d1:
 - 'How does the laboratory perform antimicrobial susceptibility testing (AST)?'
- In addition, to demonstrate that susceptibility testing is performed using up-to-date SOPs, sites must answer 'Current EUCAST standards (<3 years old)' or 'Current CLSI standards (<3 years old)' to question 3d2.1:
 - 'Which standard is used for AST?'

Extended

- To achieve 'Extended', sites must use disc diffusion methods according to SOPs for all species; this may include e-tests or broth dilution methods.
- As well as the requirements for 'Core', sites must also provide any answer other than 'Not applicable' to question 3d2:
 - 'Is the laboratory able to perform minimum inhibitory concentration (MIC) testing?'
- This builds upon the 'Core' requirements by also demonstrating that e-tests or broth dilution methods are used.

Advanced

- To achieve 'Advanced', sites must use automated identification methods (e.g. VITEK).
- Therefore, sites must answer 'Automated method (e.g. Vitek II, BD Phoenix, Microscan/Autoscan)' to question 3d1.

Question	Core	Extended	Advanced
3d1	Disc diffusion/MIC testing/Automated method (e.g. Vitek II, BD Phoenix, Microscan/Autoscan)	Disc diffusion/MIC testing	Automated method (e.g. Vitek II, BD Phoenix, Microscan/Autoscan)
3d2	Not applicable	By Etest/By broth dilution/Using both Etest and broth dilution/Other (e.g., automated methods)	By Etest/By broth dilution/Using both Etest and broth dilution/Other (e.g., automated methods)
3d2.1	Current EUCAST/CLSI standards (<3 years old)	Current EUCAST/CLSI standards (<3 years old)	Current EUCAST/CLSI standards (<3 years old)

Subcomponent 2e: Training and QA - Questions 3e1-3e2 and 2a3

Core

- To achieve 'Core', sites must perform routine training for SOPs, quality control and internal QA.
- Therefore, sites must answer 'All staff' to question 3e1:
 - 'Have relevant laboratory staff received up to date training on the sample, organism and AST SOPs for the site?'

Extended

- To achieve 'Extended', sites must take part in external quality assessment.
- As well as the requirements for 'Core', sites must also answer 'Yes' to the previous question regarding EQA, question 2a3:
 - 'Is the laboratory participating in an accredited External Quality Assessment scheme for bacterial identification and antimicrobial susceptibility testing?'

Advanced

- To achieve 'Advanced', sites must function as a regional training centre.
- Therefore, as well as the requirements for 'Extended', sites must also answer 'Yes' to question 3e2:
 - 'Does the laboratory provide training in bacterial identification/AST to other sites?'

Question	Core	Extended	Advanced
3e1	All staff	All staff	All staff
2a3	No/Not applicable	Yes	Yes
3e2	No/Not applicable	No/Not applicable	Yes

Component 3: Isolate storage (local) and referral to AMR laboratory

Subcomponent 3a: Storage of isolates - Questions 4a1-4a4

Core

- To achieve 'Core', sites must perform freezer storage (-20°C) of resistant isolates with linkage to a paper or electronic database.
- Therefore, sites must answer 'In -20°C freezers' to question 4a1:
 - 'How are resistant isolates stored?'
 - In addition, 'On agar slopes / agar stabs' is also considered acceptable.
- Furthermore, sites also must answer 'Paper-based record / log book', 'Excel', 'Formal biorepository software' or 'LIMS' to question 4a4:
 - 'Does the site maintain an up to date inventory of all the isolates stored?'

Extended

- To achieve 'Extended', sites must have reliable (generator back-up) -80°C freezer storage of resistant isolates with linkage to electronic database.
- Therefore, sites must answer 'In -80°C freezers' or 'Freeze dried' to question 4a1.
- In addition, they must also answer 'Excel', 'Formal biorepository software' or 'LIMS' to question 4a4.
 - These are all examples of electronic databases.
- Sites must also answer 'Yes' to questions 4a2 and 4a3:
 - 4a2: 'If freezers are used, is the power supply reliable with a backup generator and UPS?'
 - 4a3: 'If freezers are used, are they under a service contract with regular external monitoring of temperature?'
- This ensures that freezers are reliable.

Question	Core	Extended
4a1	In -20°C freezers/ On agar slopes / agar stabs	In -80°C freezers/Freeze dried
4a2	No/Not applicable	Yes
4a3	No/Not applicable	Yes
4a4	Paper-based record / log book/Excel/Formal biorepository software/LIMS	Excel/Formal biorepository software/LIMS

Subcomponent 3b: Transport to AMR laboratory - Questions 4b1-4b2

Core

- To achieve 'Core', sites must transfer invasive isolates to a reference AMR laboratory annually according to SOPs.
- Therefore, sites must answer 'Yes' to question 4b1:
 - 'Are invasive isolates regularly transferred to the AMR reference/coordinating laboratory for biobanking?'
- Sites must also answer 'Once a year' or 'Every 3-6 months' to question 4b2:
 - 'If yes, how frequently are isolates transferred?'

Extended

- To achieve 'Extended', sites must transfer invasive isolates to a reference AMR laboratory quarterly according to international standards for biosafety.
- Therefore, as well as 'Yes' to question 4b1, sites must also answer 'Every 1-3 months' to question 4b2.

Question	Core	Extended
4b1	Yes	Yes
4b2	Once a year/Every 3-6 months	Every 1-3 months

Subcomponent 3c: Training and QA - Questions 4c1-4c2

Core

- To achieve 'Core', sites must have routine training for isolate storage, SOPs, quality control and internal quality assessment.
- Sites must therefore answer 'Yes' to questions 4c1 and 4c2:
 - 4c1: 'Does the site have SOPs for sample referral and transfer to the AMR reference/coordinating laboratory?'
 - 4c2: 'Have relevant laboratory staff received up to date training on these SOPs?'

Extended

- To achieve 'Extended', sites must take part in an external quality assessment scheme.
- Therefore, as well as the requirements for 'Core', sites must once again have answered 'Yes' to question 2a3:
 - 'Is the laboratory participating in an accredited External Quality Assessment scheme for bacterial identification and antimicrobial susceptibility testing?'

Question	Core	Extended
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4c1	Yes	Yes
4c2	Yes	Yes
2a3	No/Not applicable	Yes

Component 4: Isolate storage (local) and referral to AMR laboratory

Subcomponent 4a: Data use - Questions 5a1-5a2 and 5a6-5a7

Core

- To achieve 'Core', anonymised individual data must be submitted to a national coordinating centre and shared regionally and internationally.
- Sites must therefore answer 'Yes, anonymised' to question 5a1:
 - 'Are AMR data submitted to a national central database (or equivalent)?'
- Sites must also answer 'Annually', 'Every 3-6 months', 'Monthly' or 'Weekly' to question 5a2:
 - 'If yes, how frequently are data submitted?'
- Finally, sites must answer 'Yes' to either question 5a6 or question 5a7:
 - 5a6: 'Are AMR data from this site shared with any local stakeholders (e.g. with stewardship committees, collaborators etc)?'
 - 5a7: 'Are AMR data from this site shared with any international stakeholders, e.g regional or global surveillance programs, research collaborators, or others?'

Advanced

- To achieve 'Advanced', sites must perform automated real time submission of data to national network.
- Therefore, as well as 'Yes, anonymised' to question 5a1, and 'Yes' to either 5a6 or 5a7, sites must also answer 'Daily automated submission' to question 5a2.

Question	Core	Advanced
5a1	Yes, anonymised	Yes, anonymised
5a2	Annually/Every 3-6 months/Monthly/Weekly	Daily automated submission
5a6 or 5a7	Yes – [local sharing group]	Yes – [local sharing group]
	Yes	Yes

Subcomponent 4b: Data linkage – Question 5b1

Core

- To achieve 'Core', clinical and laboratory data must be linked by recording both on the same lab request form.
- Sites must therefore answer 'Paper linkage (e.g. clinical data recorded on archived sample request/reporting form)' to question 5b1:
 - 'How are clinical/demographic data linked to the laboratory AMR results?'

Extended

- To achieve 'Extended', sites must have automated linkage between clinical request data and laboratory data.
- Therefore, sites must answer 'Clinical and laboratory databases use same unique patient identifier and data can be retrieved by request to IT department' to question 5b1.

Advanced

- To achieve 'Advanced', sites must have automated linkage between clinical and laboratory databases.
- Therefore, sites must answer 'Automated linkage of laboratory and clinical databases' to question 5b1.

Question	Core	Extended	Advanced
5b1	Paper linkage (e.g. clinical data recorded on archived sample request/reporting form)	Clinical and laboratory databases use same unique patient identifier and data can be retrieved by request to IT department	Automated linkage of laboratory and clinical databases

Subcomponent 4c: Data governance – Question 5c1

Core

- To achieve 'Core', sites must have a data sharing policy and agreements in place in collaboration with the Ministry of Health and/or a national public health institute.
- Sites must therefore answer 'Yes' to question 5c1:
 - 'Does the site have a formal AMR data sharing policy or agreement in place with the ministry of health and/or national public health institute?'

Question	Core
5c1	Yes

Footnotes:*** Subcomponent 1c:**

In phase 1 (before 2024) of the Fleming Fund, question 1c1 did not exist. Instead, 1c2 and 1c3 were used to determine the level of function.

- To achieve 'Core', sites must answer 'Yes' to question 1c2:
 - 'Are clinical details recorded on requests to the microbiology laboratory?'
- Sites must also answer either 'Chosen from a checklist of syndromes' or 'Free text' to question 1c3:
 - 'If yes, how are the details recorded?'
- To achieve 'Extended', as well as answering 'Yes' to question 1c2, they must also answer 'Recorded as a standardised case definition (e.g. ICD-11)' to question 1c3.

**** Subcomponent 2c:**

In 2024, new priority pathogens were added to the 'WHO GLASS Manual for Resistance Surveillance'. Given that the roadmap requires 'relevant priority pathogens', sites were therefore also required to identify these additional pathogens for 'Core' level function, from 2024 onwards.

Supplementary Figures:

AMR surveillance component		Requirements and standards for core level	Extended level activities	Advanced level activities
Overall aim		Surveillance data inform individual care	Surveillance data drive local and national policy (e.g. empiric treatment guidelines, drug procurement) and public health activities	
Clinical admission assessment and investigation	Clinical admission assessment	Clinical history and examination and investigation based on physician (syndromic) diagnosis.	Systematic clinical history and examination according to clinical algorithms in all patients presenting with suspected infection.	Standardised admission proforma documenting clinical signs and symptoms used to guide diagnosis.
	Clinical data	Clinical data included in (paper) request for laboratory investigation, with unique alphanumeric identifier	Clinical data included in (electronic) request for laboratory investigation, with unique alphanumeric identifier	Linkage of extended clinical data (e.g. vital signs, blood results, outcomes) with laboratory data
	Clinical investigation	Systematic investigation based on physician syndromic diagnosis	Systematic investigation based on clinical findings.	
	Training and quality assurance	Routine training for surveillance SOPs, quality control and Internal Quality Assessment.	External Quality Assessment	Functions as a regional training centre
Isolate identification and susceptibility testing	Sample transport	Samples transported according to local SOPs	Samples transported according to international biosafety standards	
	Sample registration	Local laboratory paper based data system	Electronic laboratory data system	
	Culture and identification	Automated blood culture system and capacity to identify the relevant priority pathogens according to SOPs	Automated blood culture; CSF, urine, stool and swab culture, identifying all isolates according to SOPs for all priority pathogens.	Automated identification (e.g. MALDI-TOF)
	Susceptibility testing	Use of disc diffusion for blood culture priority pathogens according to SOPs	Use of disc diffusion methods according to SOPs for all species; may include e-tests or broth dilution methods.	Automated identification (e.g. VITEK)
	Training and QA	Routine training for SOPs, quality control and internal QA	External quality assessment	Functions as a regional training centre
Isolate storage (local) and referral to AMR laboratory	Storage of isolates	Freezer storage (-20°C) of resistant isolates with linkage to paper or electronic database	Reliable (generator back-up) freezer storage (-80°C) of resistant isolates with linkage to electronic database	
	Transport to AMR laboratory	Invasive isolates are transferred to AMR laboratory annually according to SOPs	Invasive isolates are transferred to AMR laboratory quarterly according to international standards for biosafety	
	Training and QA	Routine training for isolate storage, SOPs, quality control and internal quality assessment	External quality assessment	
Data review	Data use	Anonymised individual data submitted to national coordinating centre and shared regionally and internationally		Automated real time submission of data to national network
	Data linkage	Clinical and laboratory data linked by recording them on the same lab request form	Automated linkage between clinical request data and laboratory data	Automated linkage between clinical and laboratory databases
	Data governance	Data sharing policy and agreements in place in collaboration with the Ministry of Health and/or national public health institute		

Supplementary Figure 1. Human health sentinel site functions for AMR surveillance (Seale, et al., 2017). Table demonstrating four essential overall AMR surveillance components which a sentinel site must be able to perform. These are further divided into 15 subcomponent functions. Sites can perform these functions at either a level of 'Core', 'Extended', or 'Advanced', and the requirements/standards to achieve each level are detailed in the table. If a site does not meet the requirements of any of these levels, the site is designated to be functioning at a 'Precore' level for that subcomponent function.